2.1 Lines and Angles

A transversal crossing two lines creates eight angles, and the names — corresponding, alternate interior, alternate exterior, same-side interior — describe exact positional relationships, not vague location. Learning to read these positions correctly (interior vs. exterior, same side vs. opposite side of the transversal) is what makes Section 2.2's parallel-line theorems almost automatic once you get there. This section pushes past simple recall by asking you to classify a pair from a description, count angles across multiple lines, and work out every relationship a single angle has with all seven others.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. A line intersecting two coplanar lines at different points is a what?

2. Matching-position angles at two intersections are called what?

3. Interior angles on opposite sides of the transversal?

4. Interior angles on the same side of the transversal?

5. Exterior angles on opposite sides of the transversal?

6. Two lines cut by a transversal form how many angles?

7. Interior angles lie where?

8. Exterior angles lie where?

9. At one intersection, opposite angles are what?

10. Adjacent angles on a straight line at an intersection form a what?

11. Top-left at first intersection and top-left at second: what pair?

12. Bottom-left interior and top-right interior: what pair?

13. Must the two lines cut by a transversal be coplanar for these angle pairs?

14. A single transversal crosses 4 distinct parallel lines. How many intersection points and how many total angles are formed?

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15. In the standard 8-angle figure, classify $\angle 6$'s relationship to $\angle 1$, and separately to $\angle 4$.

✓ Answer Key

1. Transversal
2. Corresponding
3. Alternate interior
4. Same-side interior
5. Alternate exterior
6. 8
7. Between the two lines
8. Outside the two lines
9. Vertical
10. Linear pair
11. Corresponding
12. Alternate interior
13. Yes
14. 4 intersection points; 16 angles total
15. $\angle 6$ & $\angle 1$: no special named relationship; $\angle 6$ & $\angle 4$: same-side interior angles